Github files scheme

Files in the folder Diet		
GITHUB FILES	**DESCRIPTION**	**OUTPUT FILES**
1_extraction_tables_from_Html_diaries.py	Convert food_diary from html format to csv or rda format	TRIESTE_data.rda/ CAGLIARI_data.rda/ BOLOGNA_data.rda
2_Identify_MyFitnessPal_Outliers.R	Computes and generate files containing: – Total mean nutrients per person – Weekly mean nutrients per person also generates Boxplot images with outliers	<sample>_Diaries_TM.csv <sample>_Diaries_WM.csv <sample>__outliers.jpeg <sample>_<trait>_outliers_per_week.jpeg
3_computing_3days_average	Computes the average of nutrients across the 3days before collection date (nutrients_3days) for each visit	TRIESTE_3day.rda CAGLIARI_3day.rda BOLOGNA_3day.rda
4_Prepare_Diaries_3day.R	Generates a file with nutrients_3day means for each hormonal phase, wide format	Diaries_3day_phases.csv

5_FFQ_convert.R	From the FFQ questionnaire and the BDA-derived food composition table (nutrients per 100g/100ml), this script computes – daily nutrients intake – daily food intake and gives a list of potential errors in the questionnaire	FFQ.csv (daily nutrients intake) FFQ_die.csv (daily food intake) warnings.txt
6_Derive_Diet_Scores.R	Generate file with aMED,HEI, DASH, AHEI scores, starting from weekly intake of each food item derived from the FFQ questionnaire (FFQ_week.csv derived from FFQ_die.csv)	Diet_Scores.csv
7_Compare_Study_Centers.R	Computes – Statistical tests to compare centers Also creates Boxplot images with pvalues	<label>_pairwise_pvalues.csv <label>_kruskal_wallis_pvalues.csv <label>_samples.jpeg
8_Diaries_Lmer.R	Computes linear and spline mixed models from nutritional intake (nutrients_3days) across hormonal phases; Input file “Diaries_3day_phases.csv” generated by *Prepare_Diaries_3day.R* Also generates figures	LMER_model_<label>.csv SPLINE_model_<label>.csv and jpeg LINEAR_model_<label>.csv and jpeg
9_Diaries_Boxplots.R	Computes and generates:	pairwise_pvalues_<label>.csv

	– Wilcoxon test across phases – Boxplots of traits across phases	<trait>_paired_pairwise_boxplot_<label>.jpeg
10_Identify_Clusters.R	Computes - PCA K-means clusters - means values per cluster	<label>_clusters.csv <label>_clusters_means.csv
11_Compare_FFQ_Diaries.R	Compares daily nutrients intakes from FFQ and Diaries (total means) with – Paired Wilcoxon Signed-Rank tests – Spearman correlations and p-values also creates plots	"FFQ_Diaries_Wilcoxcon_pvalues.csv (+"FFQ_Diaries_means.jpeg) "FFQ_Diaries_Spearman_pvalues.csv (+"FFQ_Diaries_correlations.jpeg)
Files in the folder Microbiome		
1.KneadData.sh	Performs trimming, quality control, and contaminant removal of paired-end reads using KneadData, and summarizes read counts	For each sample <sample>: <outDIR>/<sample>/<sample>_kneaddata.fastq Cleaned and decontaminated reads <outDIR>/<sample>/<sample>_kneaddata.log Processing log file <outDIR>/<sample>/*fastqc_start/ FastQC reports on raw reads <outDIR>/<sample>/*fastqc_end/ FastQC reports on cleaned reads
2.MetaPhlAn.sh	Performs taxonomic profiling of shotgun metagenomic samples using MetaPhlAn.	For each sample <sample>: <outDIR>/<sample>/<sample>_metaphlan.txt
3.Humann.sh	Performs functional profiling of shotgun	# Per-sample outputs: $outDIR/<sample>/<sample>_humann_genefamilies.tsv $outDIR/<sample>/<sample>_humann_pathabundance.tsv

	metagenomic samples using HUMAnN.	$outDIR/<sample>/<sample>_humann.log # Regrouped gene families (EC level 4): $outDIR/<sample>/<sample>_humann_genefamilies.uniref90_level4ec.tsv # Renormalized (relative abundance, community level): $mergedOUTdir/temp/<sample>_humann_genefamilies.uniref90_level4ec.community.tsv # FINAL MERGED OUTPUT: $mergedOUTdir/MGS_FECAL_humann_genefamilies_uniref90_level4ec.community.tsv
4.Data_Preparation.R	Integrates metadata, sequencing statistics, phenotypes, and dietary information	species.MGS_FECAL_merged_abundance_table.txt species.MGS_FECAL_merged_abundance_table.rds species.MGS_FECAL_merged_abundance_table_clr.txt species.MGS_FECAL_merged_abundance_table_clr.rds pathways.MGS_FECAL_merged_abundance_table_clr.txt pathways.MGS_FECAL_merged_abundance_table_clr.rds genefam.MGS_FECAL_merged_abundance_table.txt genefam.MGS_FECAL_merged_abundance_table.rds genefam.MGS_FECAL_merged_abundance_table_clr.txt genefam.MGS_FECAL_merged_abundance_table_clr.rds
5.Alpha_diversity_plots.R	Calculates Shannon alpha diversity (species level) and visualizes its distribution across phases	Alpha_Diversity_phase.pdf
6.Beta_diversity_plots.R	Computes beta diversity from species-level abundances using CLR transformation, adjusts for technical covariates, and visualizes community differences	Beta_Diversity_phase.pdf PCOA_Beta_Diversity_phase.pdf
7.Wilcoxon_tests.R	Performs paired Wilcoxon signed-rank tests across phases for microbiome	Wilcoxon_results_beta.csv Wilcoxon_results_alpha.csv

	features (species or gene families) and beta diversity and alpha diversity	
8.LMM_Associations.R	Performs linear mixed models to assess associations between microbiome features and phase, hormones, and covariates, accounting for repeated measures	phase_associations.txt PROG_associations.txt PRL_associations.txt LH_associations.txt FSH_associations.txt each_macronutrients_associations.txt each_FFQ_food_associations.txt
9.Heatmap_plot.R	Generates heatmaps from association results (e.g., LMM outputs), displaying effect sizes and highlighting significant features	phase_heatmap.txt PROG_heatmap.txt PRL_heatmap.txt LH_heatmap.txt FSH_heatmap.txt each_macronutrients_heatmap.txt each_FFQ_food_heatmap.txt
10.Forest_plot.R	Generates forest plots of effect coefficients with 95% confidence intervals for feature–phase associations across multiple models	phase_forest_plot.txt PROG_forest_plot.txt PRL_forest_plot.txt LH_forest_plot.txt FSH_forest_plot.txt each_macronutrients_forest_plot.txt each_FFQ_food_forest_plot.txt
11.Permanova.R	Performs beta diversity analysis using CLR-transformed species data and tests associations with metadata using PERMANOVA (adonis2)	Phase_PERMANOVA_univariable.csv PROG_PERMANOVA_univariable.csv PRL_PERMANOVA_univariable.csv LH_PERMANOVA_univariable.csv

		FSH_PERMANOVA_univariable.csv each_macronutrients_PERMANOVA_univariable.csv each_FFQ_food_PERMANOVA_univariable.csv PERMANOVA_multivariable.csv
12.Mediation.R	Performs mediation analysis to assess whether microbiome features mediate the relationship between exposure variables and outcomes	mediation_results.csv
13.Pcoa_correlation.R	Computes correlations between PCoA axes (beta diversity) and FSH levels stratified by phase, and generates multi-panel plots	PCOA_FSH_correlations.pdf
14.Species_composition_pie.R	Visualizes microbiome compositional differences across phases using mean relative abundance and pie charts	composition_pies.pdf
15.Gene_families_composition_bar.R	Visualizes compositional profiles of gene families across samples and phases using relative abundance stacked bar plots	gene_family_composition.pdf
16.Gene family_boxplot.R	Visualizes CLR-transformed gene family abundances across phases using violin + boxplots	genefam_boxplots.pdf